

ARISTO EDUCATIONAL PRESS
HONG KONG DIPLOMA OF SECONDARY
EDUCATION EXAMINATION

(Mock Test 30)

Suggested Solution

Paper 1

- | | | |
|-------|-------|-------|
| 1. C | 11. C | 21. C |
| 2. D | 12. B | 22. B |
| 3. A | 13. D | 23. C |
| 4. B | 14. A | 24. B |
| 5. B | 15. B | 25. A |
| 6. A | 16. B | 26. A |
| 7. D | 17. C | 27. A |
| 8. D | 18. D | 28. D |
| 9. C | 19. C | 29. C |
| 10. A | 20. D | 30. B |

Explanation

1. Answer: C

The opportunity cost of continuing to use the computer is the value of the highest-valued option forgone, i.e. the value of reselling the computer (\$2,000). The repairing cost of \$3,000 is the monetary cost of continuing to use the computer. The cost of continuing to use the computer = \$2,000 + \$3,000 = \$5,000.

2. Answer: D

Karson would choose the first option as the benefit of choosing the first option (the value of the first option) is higher than the cost (the value of the second option).

Option D is correct. This would reduce the value of the third option. This would not affect Karson's cost or benefit of choosing the first option.

Option A is incorrect. This would reduce the value of the first option, thus reducing Karson's benefit of choosing the first option.

Option B is incorrect. This would raise the value of the second option, thus raising Karson's cost of choosing the first option.

Option C is incorrect. This may increase the value of the third option to a level higher than that of the second option, so the cost of choosing the first option may change from the value of the second option to that of the third option.

3. Answer: A

Provision of the delivery service involves the use of scarce resources, such as drivers, trucks and gasoline, which have alternative uses. The marginal cost of providing the delivery service is positive. The delivery service is thus an economic good.

Option B is incorrect. Both free goods and economic goods can satisfy human wants.

Option C is incorrect. Goods that are free of charge are not necessarily free goods. Economic goods can be provided to consumers free of charge.

Option D is incorrect. More of the free delivery service is preferred.

4. Answer: B

(2) is incorrect. The government decides what resources (an electronic platform) to use for processing admission applications, which involves a change in the method of production.

5. Answer: B

Option A is incorrect. A private limited company can issue shares and bonds to raise capital while a sole proprietorship cannot, so a sole proprietorship does not have a wider source of capital than a private limited company.

Option C is incorrect. A private limited company is not required to disclose its accounting information to the public either.

Option D is incorrect. A sole proprietorship is not a legal entity, implying that Cyrus bears unlimited liability and he may have to use personal property to settle the firm's outstanding debts. This would not be a reason for Cyrus's setting up a sole proprietorship.

6. Answer: A

As the number of owners of firm A is more than one and it lacks lasting continuity, firm A is a partnership. As firm B has more than 50 owners and there is lasting continuity, firm B is a public limited company.

Option A is correct. Firm B (a public limited company) must disclose its accounting information to the public regularly while firm A (a partnership) need not do so.

Option B is incorrect. Firm A (a partnership) is not a legal entity and its owners bear unlimited liability. Firm B (a public limited company) is a legal entity and its owners enjoy limited liability.

Option C is incorrect. The shares of firm B (a public limited company) can be transferred without the consent of other shareholders while the ownership of firm A (a partnership) can only be transferred with the consent of all partners, so firm B has a higher risk of being taken over.

Option D is incorrect. Public limited companies are not necessarily more profitable than partnerships.

7. Answer: D

8. Answer: D

Simple division of labour is when a worker specialises in producing a particular good while complex division of labour is when a worker specialises in a particular stage of the production of a good. Each employee is involved in a part of the process of video production, so this is a complex division of labour.

9. Answer: C

Sunlight is a natural resource which is a gift of nature. Natural resources become capital after going through man-made processes. Collected solar energy is capital. However, sunlight itself belongs to land.

10. Answer: A

(1) is correct. Talents are well-educated and experienced. Attracting more talents to work in Hong Kong will increase the average labour productivity of Hong Kong.

(2) is correct. The scheme raises the ability of graduates from designated universities to move to work in Hong Kong from other places, their geographical mobility will increase.

(3) is incorrect. The scheme does not affect the ability or willingness of Hong Kong workers to change from one occupation to another.

11. Answer: C

Average labour productivity in year 1 = $600 \text{ units of output} / (15 \times 8 \text{ man-hours})$
= 5 units of output per man-hour

Average labour productivity in year 2 = $675 \text{ units output} / (20 \times 7.5 \text{ man-hours})$
= 4.5 units of output per man-hour

The firm's average labour productivity decreased.

If the firm does not provide adequate training for new workers, average labour productivity may decrease.

Option A is incorrect. If worker morale improves, average labour productivity will increase.

Option B is incorrect. If the firm uses more capital goods, workers can produce more output with the same number of working hours, and average labour productivity will increase.

Option D is incorrect. If the wage payment method is changed from time rate to piece rate, workers' incentive to work will increase and average labour productivity will increase.

12. Answer: B

Machinery (units)	Labour (units)	Marginal product of labour (units)	Total product (units)	Average output (units)
5	1	4	4	4
5	2	6	10	5
5	3	8	18	6
5	4	10	28	7
5	5	2	30	6

13. Answer: D

Output (units)	2	3	4	5
Average variable cost (\$)	3	4	5	6
Total variable cost (\$)	6	12	20	30
Marginal cost (\$)	-	6	8	10
Total fixed cost (\$)	8	8	8	8
Total cost (\$)	14	20	28	38

At the profit-maximising output level (4 units), the marginal revenue equals the marginal cost (= \$8).

Profit = Total revenue – Total cost = $(\$8 \times 4) - \$28 = \$4$

Producer surplus = Total revenue – Total variable cost = $(\$8 \times 4) - \$20 = \$12$

14. Answer: A

When the cost of importing leather increases, the marginal cost of producing leather products will increase and the profit-maximising output will decrease.

Option B is incorrect. When the rental of the store increases, the fixed cost of producing leather products will increase and the profit-maximising output will not be affected.

Option C is incorrect. When leather products become more popular, demand for leather products and the market price will increase. The profit-maximising output will increase.

Option D is incorrect. When the wage rate of part-time workers decreases, the marginal cost of producing leather products will decrease, and the profit-maximising output will increase.

15. Answer: B

(1) is correct. As the price of white-plate sushi increases, its quantity demanded will decrease.

(2) is correct. After the price of all dishes is increased by \$2, the relative price of white-plate sushi (in terms of black-plate sushi) will increase. According to the law of demand, people will consume less white-plate sushi relative to black-plate sushi. The ratio of white-plate sushi sold to black-plate sushi sold will thus decrease.

(3) is incorrect. The change in total revenue of the restaurant will be uncertain, depending on the price elasticity of demand for sushi.

16. Answer: B

When the interest rate on vehicle loans falls, the cost of buying a car will decrease and the demand for second-hand cars will increase.

Option A is incorrect. When the price of second-hand cars is expected to decrease, demand for second-hand cars will decrease.

Option C is incorrect. New cars and second-hand cars are substitutes. When the price of new cars decreases, demand for second-hand cars will decrease.

Option D is incorrect. When more second-hand cars are scrapped, the supply of second-hand cars will decrease while there will be no effect on demand.

17. Answer: C

A decrease in the price of fuel for aircraft will increase the supply of air services, causing the equilibrium position to change from A to B.

Option A is incorrect. This will increase the demand for air services between Hong Kong and Japan.

Option B is incorrect. This will reduce the demand for air services between Hong Kong and Japan.

Option D is incorrect. This will reduce the supply of air services between Hong Kong and Japan.

18. Answer: D

When the price of cattle feed falls, the cost of producing beef will decrease and the supply of beef will increase. At the same time, when the production cost of plant-based meat decreases, the supply of plant-based meat will increase and the price will decrease. Plant-based meat and beef are substitutes. The decrease in the price of plant-based meat will reduce the demand for beef.

19. Answer: C

Hair stylists are a factor of production that provides hairdressing services. The demand for hair stylists can be regarded as a derived demand for hairdressing services.

Option A is incorrect. Flour is a raw material for making bread. The demand for flour can be regarded as a derived demand for bread.

Option B is incorrect. Public healthcare services and private healthcare services can satisfy similar wants or needs, so they are in competitive demand.

Option D is incorrect. Mobile network services and smartphones are usually used together to satisfy the need for mobile services, so they are in joint demand.

20. Answer: D

Given that good X and good Y are in joint supply. When the price of good Y decreases, the supply of good X will decrease. The price will increase and the quantity transacted will decrease.

Percentage decrease in quantity transacted = $(190 - 210) / [(210 + 190) / 2] \times 100\% = 10\%$

As the percentage change in quantity demanded (10%) is smaller than the percentage change in price (15%), the demand for good X is inelastic.

If good X is a habit-forming good, the demand for good X tends to be less elastic.

21. Answer: C

Demand for private medical services increases. Given an upward-sloping supply curve, both the equilibrium price and quantity will increase. The supply is elastic because the percentage change in quantity supplied (10%) is greater than the percentage change in price (5%).

22. Answer: B

Before the imposition of the price ceiling, the price of good X is \$10 and the quantity transacted is 20 units.

After the imposition of the price ceiling, the price of good X is \$6 and the quantity transacted is 20 units.

Option B is correct. The original total expenditure is \$200 = (20 units × \$10) and the new total expenditure is \$120 (= 20 units × \$6), so the total expenditure decreases by \$80.

Option A is incorrect. The quantity transacted will remain at 20 units.

Option C is incorrect. When the price is \$6, the quantity demanded (28 units) will be greater than the quantity supplied (20 units). There will be a shortage of good X.

Option D is incorrect. When the price of good X decreases, the demand for its complements will increase.

23. Answer: C

The original equilibrium price is \$4 and the equilibrium quantity is 16 units. After the increase in demand, the new equilibrium price is \$8 and the equilibrium quantity is 18 units.

At the equilibrium point, the quantity demanded is equal to the quantity supplied. Without market intervention, the original quantity supplied is 16 units and the new quantity supplied is 18 units.

(1) is correct. If the government imposes a quota of 16 units, the quota is effective (the quota limit is below the equilibrium quantity). The quantity transacted will be 16 units and the market price will be \$10.

(2) is correct. The price floor of \$10 is above the equilibrium price (\$8), so the price floor is effective. Under the price floor, the quantity transacted will be equal to the quantity demanded, i.e. 16 units.

(3) is correct. If the government imposes a per-unit tax of \$6 on good X, the supply of good X will decrease. The new equilibrium price is \$10 and the quantity transacted is 16 units.

Unit price (\$)	Quantity demanded (units)		Quantity supplied (units)	
	Before an increase in demand	After an increase in demand	Without market intervention	With a per-unit tax of \$6
10	10	16	-	16
8	12	18	18	-
6	14	20	-	-
4	16	22	16	-
2	18	24	-	-

(4) is incorrect. If the government provides a per-unit subsidy of \$4 to the producers of good X, the supply of good X will increase and the quantity transacted of good X will further increase.

24. Answer: B

Option A is incorrect. Firms under oligopoly engage in non-price competition, but this is not sufficient to support the answer because firms of other imperfectly competitive markets (e.g. monopolistic competition) also engage in non-price competition.

25. Answer: A

The original consumer surplus is equal to area (V + W + X) and the new consumer surplus is equal to area V. The consumer surplus is reduced by area (W + X).

The original total social surplus is equal to area (V + W + X + Y + Z) and the new total social surplus is equal to area (V + W + Y). The total social surplus is reduced by area (X + Z).

26. Answer: A

After the imposition of the quota of 2 units, the market price will be \$9.

Consumer surplus after the quota is imposed = $(\$10 - \$9) + (\$9 - \$9) = \$1$

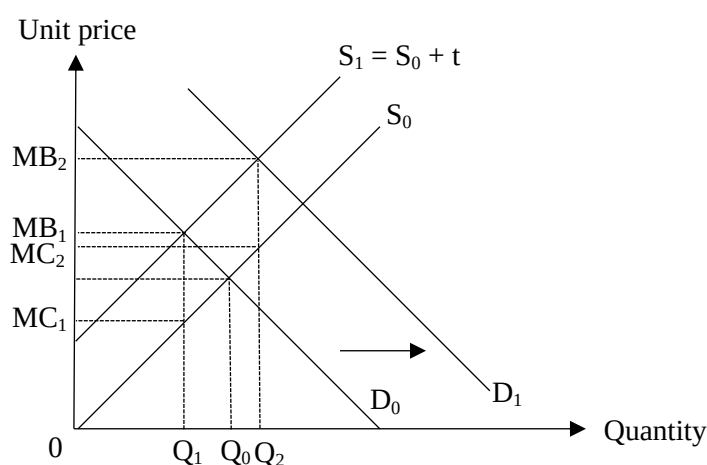
Producer surplus after the quota is imposed = $(\$9 - \$4) + (\$9 - \$5) = \$9$

27. Answer: A

(1) is correct. After the increase in demand, the price and quantity transacted of good X will increase while the marginal cost of good X will remain unchanged, so the producer surplus will increase.

(2) is correct. After the increase in demand, the quantity transacted of good X will increase, while the per-unit tax will remain unchanged, so the total tax revenue will increase.

(3) is incorrect. When the government imposes a per-unit tax on good X, the difference between marginal benefit and marginal cost is equal to the amount of the per-unit tax. After the increase in demand, the per-unit tax remains unchanged, so the gap between marginal benefit and marginal cost will also remain unchanged.



28. Answer: D

Fining the restaurant will increase the restaurant's marginal private cost so that the external cost can be internalised. The gap between marginal social cost and marginal social benefit will be narrowed and efficiency will improve.

Option A is incorrect. Which solution should be adopted depends on many factors, such as the cost of the solution. There is not enough information to determine which solution should be adopted.

Option B is incorrect. Fining the restaurant may improve efficiency, but efficiency may not be restored. If the fine cannot accurately reflect the external cost, social optimality will not be reached.

Option C is incorrect. Other than fining the restaurant, there are other market solutions. For example, the residents living nearby can pay the restaurant to improve its sanitation.

29. Answer: C

Country A's Gini coefficient based on original monthly household income has always been higher than the Gini coefficient based on post-tax monthly household income, reflecting that its tax system can help improve income inequality. A progressive tax system can help improve income inequality.

Option A is incorrect. The diagram does not reflect the income of households.

Option B is incorrect. The Gini coefficient based on post-tax monthly household income does not reflect the effect of transfer payments on income distribution.

Option D is incorrect. The difference between the Gini coefficient based on original monthly household income and the Gini coefficient based on post-tax monthly household income does not reflect the amount of tax payment.

30. Answer: B

(1) is correct. Students will not be deprived of education because of the financial difficulties of their families. Education will also help improve their productivity and hence income-earning power in the future.

(3) is correct. Providing an allowance to employers who employ people with disabilities can increase the chances of people with disabilities being employed and thus help increase their income.

(2) is incorrect. When the government distributes consumption vouchers to all people aged 18 or above, compared with high-income earners, low-income earners do not get more resources to improve their productivity.

Paper 2

Section A (30 marks)

1. (a) Yes. The lower-than-expected output after the expansion of the garment factory will only decrease the value of the chosen option, while the highest-valued option forgone will not be affected. (2 marks)
(b) No. More local garment factories switching from imported fabric to local fabric may cause the sales of the fabric supplier to increase, and this may increase the value of the third option to a level higher than that of the second option. The cost of expanding the existing garment factory may change from the value of the second option (i.e. acquiring a boutique) to the value of the third option (i.e. acquiring a fabric supplier). (3 marks)
2. (a) The private property rights over these residential properties are weakened because the owners' right to transfer (or sell) their residential properties to non-Canadians is confined. (2 marks)
(b) Positive statement. It describes what is, with no value judgement. It is refutable by facts. (2 marks)
3. (a) Feeding feral pigeons can lead to congregations of feral pigeons, increasing the chances of other people being exposed to harmful germs. People who feed feral pigeons impose a cost on other parties but they do not give any compensation for this extra cost. A negative externality thus exists and marginal social cost is greater than marginal social benefit. The quantity of feral pigeon feeding behaviour is higher than the efficient level. (3 marks)
(b) The government can raise the penalty for feeding feral pigeons so as to increase the marginal private cost of feeding feral pigeons. The quantity of feral pigeon feeding behaviour will decrease towards the efficient level. (2 marks)
OR
Any reasonable answer (2 marks)
- 4.

Labour (units)	Machines (units)	Average product of labour (units)	Total product (units)	Marginal product (units)
1	8	10	10	10
2	8	11	22	12
3	8	13	39	17
4	8	15	60	21
5	8	16	80	20

The law of diminishing marginal returns states that in the short run, as more units of a variable factor are added to a given quantity of fixed factor successively, marginal product will decrease eventually, other things being constant. (3 marks)

Yes. When labour increases from 4 to 5 units, the marginal product of labour decreases from 21 to 20 units, so the law of diminishing marginal returns holds. (3 marks)

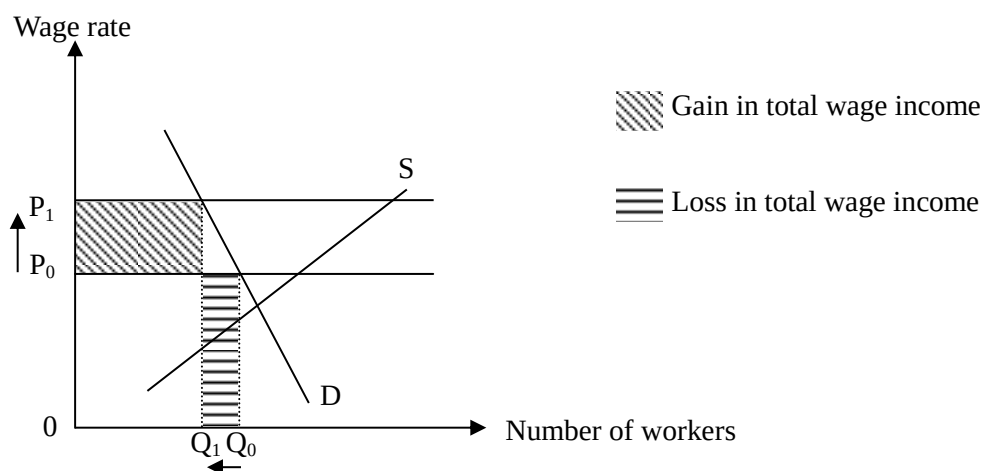
5. When the same amount of fare is added to each trip, the relative price of long-distance trips in terms of short-distance trips would decrease. The law of demand implies that people would take more long-distance trips relative to short-distance trips. The proportion of passengers travelling long distance would increase. The average travel distance of taxi passengers would increase. (4 marks)

6. Verbal elaboration:

- When the statutory minimum wage rate is raised, the wage rate will increase and the number of workers will decrease. (1 mark)
- If the demand for workers is inelastic, the percentage increase in the wage rate will be larger than the percentage decrease in the number of workers. The total wage income of employed workers will increase. (2 marks)

Illustrate in the diagram:

- minimum wage rate increases (1 mark)
- correct positions of gain in total wage income and loss in total wage income (1 mark)
- gain in total wage income > loss in total wage income (1 mark)



Section B (40 marks)

7. (a) Tertiary production, (1 mark)
because estate agents provide estate agents' services. (1 mark)
- (b) An advantage:
- The estate agency companies can share some business risks with the estate agents. When the income of the estate agency companies decreases, they would make lower payments to the estate agents.
 - The estate agency companies can have a lower cost of monitoring the performance of estate agents as they will have a higher work incentive.
 - Any reasonable answer
- (Mark the **FIRST** point only)
- (c) A possible demand-side factor:
- There is a decrease in the number of property transactions, so the demand for estate agents' services decreases. This leads to a decrease in demand for estate agents.
 - Any reasonable answer
- (Mark the **FIRST** point only)
- (d) The occupational mobility of labour will increase. (1 mark)
Waiving the application fees for estate agents to obtain licenses will lower the cost of becoming an estate agent. (2 marks)
8. (a) An example of variable cost:
- wages for part-time employees
 - cost of food ingredients
 - any reasonable answer
- (Mark the **FIRST** point only)
- (b) When some restaurants are closed during the Lunar New Year period, the number of substitutes for a restaurant's services would decrease. The price elasticity of demand for restaurant services would decrease. (2 marks)
- (c) 'For whom to produce', (1 mark)
because the change in prices involves how to distribute goods and services to people in society.
When the menu prices are increased, those who are willing and able to pay higher prices can get the service. (1 mark)

(d) Verbal elaboration:

- Demand for the restaurant services would increase.

(1 mark)

- The quantity transacted would increase.

(1 mark)

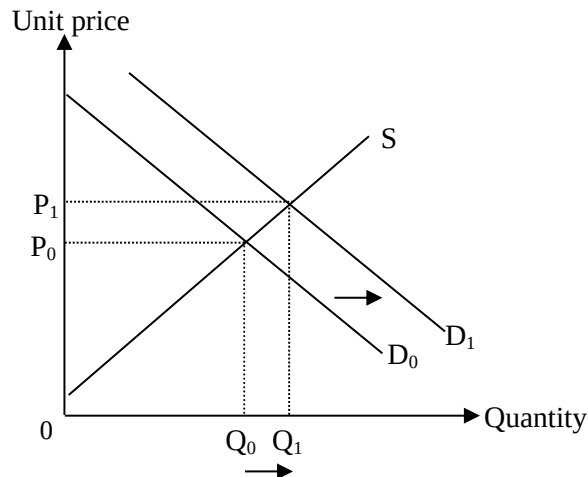
Illustrate in the diagram:

- rightward shift of the demand curve

(1 mark)

- quantity transacted increases

(1 mark)



9. (a) Natural gas used to produce industrial products/to generate electricity/for export is a producer good.

(1 mark)

(b) Possible reasons:

- The demand for natural gas is different in different parts of Australia.
- The cost of producing natural gas is different in different parts of Australia.
- Any reasonable answers

2@, max: 4 marks

(Mark the **FIRST TWO** points only)

(c) Verbal elaboration:

Natural gas is a raw material for the production of industrial products. When the price of natural gas increases, the production cost of industrial product manufacturers would increase. (1 mark)

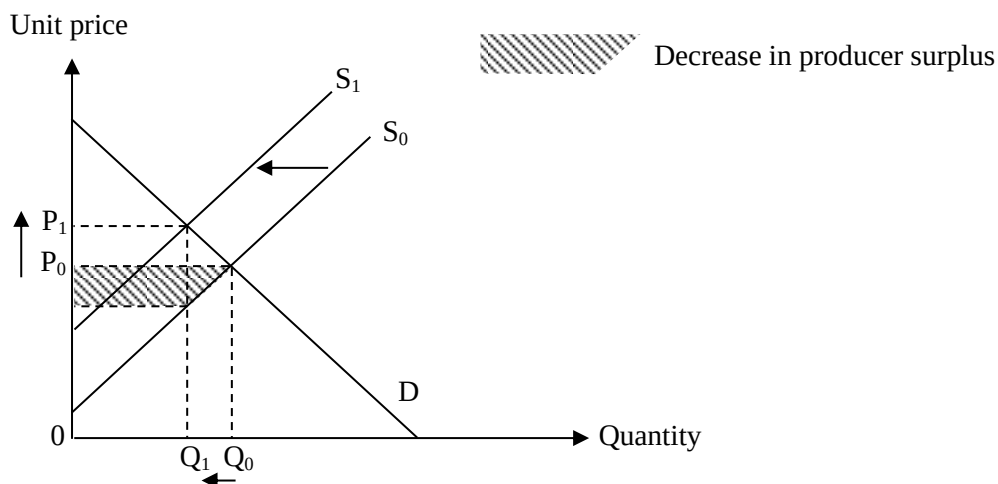
The supply of industrial products would decrease. (1 mark)

Producer surplus would decrease. (1 mark)

Illustrate in the diagram:

- leftward shift of the supply curve (1 mark)

- correct position of the decrease in producer surplus (1 mark)



(d) Marks award for effective communication (EC: max 2 marks)

Marks	Performance
2	● Arguments supported with the source/data and appropriate economic theories ● Relevant materials presented ● Well-organised and coherent answers presented without repetition of ideas ● Ideas clearly and fluently expressed with appropriate use of language/words/terms/symbols
1	● Arguments presented with some support of the source/data and economic theories ● Some irrelevant materials presented ● Answers presented in a less organised way with some repetition of ideas ● A clear message conveyed with some inappropriate use of language/words/terms/symbols
0	● Arguments presented with no support of the source/data and economic theories ● Materials unrelated to the gist of the question presented ● Inconsistent arguments presented ● Limited ideas expressed with inappropriate use of words/terms/symbols

The maximum marks for content are 10 marks. Answers may include the following:

Exporters of Australian natural gas

- Natural gas for export and natural gas for domestic use are in competitive supply. Imposing an effective price ceiling will lower the price of domestic natural gas. Producers of natural gas will tend to sell natural gas to exporters rather than to domestic buyers. For exporters of natural gas, the supply of natural gas for export will increase. Assuming the supply curve shifts in a parallel manner, producer surplus will increase.
- When the export of natural gas is restricted, the quantity of natural gas exported will decrease and the price will increase. The change in producer surplus for exporters will be uncertain.

Domestic buyers of Australian natural gas

- Imposing an effective price ceiling will lead to excess demand/a shortage in the domestic natural gas market. Non-price competition will arise. Some domestic buyers may not be able to obtain the quantity of natural gas they plan to buy. Although the monetary cost paid by domestic buyers who succeed in buying natural gas will decrease, the non-monetary cost may increase. The full cost may increase.
- Restricting the export of natural gas will increase the domestic supply of natural gas. The price of domestic natural gas will decrease and the consumer surplus of domestic buyers will increase.
- The same amount of expenditure on natural gas takes a larger percentage of the income of low-income households than that of high-income households. Restricting the export of natural gas will reduce the price of domestic natural gas, which can reduce income inequality.

Carbon emissions in Australia

- An effective price ceiling will lead to excess demand/a shortage in the domestic natural gas market. Energy companies may not be able to purchase enough natural gas for electricity generation. Australia's carbon emissions will increase if energy companies use more petroleum and coal to generate electricity.
- Natural gas, petroleum and coal can be used to generate electricity and they are substitutes. Restricting the export of natural gas will increase the domestic supply of natural gas, which will lower the domestic price of natural gas. Assume the prices of petroleum and coal remain unchanged, energy companies will tend to use natural gas for electricity generation. Australia's carbon emissions will decrease.